Poisson regression

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Warmup

Work on the warmup activity (handout)

Warmup

$e^{0.0265} \approx 1.027$

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	0.276493	0.087294	3.167	0.00154	**
phd	0.008071	0.026358	0.306	0.75944	
ment	0.026537	0.002001	13.264	< 2e-16	***
kid5	-0.099696	0.035248	-2.828	0.00468	**

How would we interpret the coefficient on the number of articles published by the student's mentor?

Each additional article published by the student's mentor is associated with an increase of 2.7% in the predicted # of articles published by the student, holding PhD prestige and # of children fixed

Warmup

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What is the predicted number of articles for a student at a PhD program with a prestige score of 4, whose mentor published 30 articles in the last 3 years, and who has no children?

$\log(\hat{\lambda}) = 0.276 + 0.008\,1(4) + 0.026\,5(30) - 0$

$\hat{\lambda} = \exp\{\quad\quad\quad\quad\}$

≈ 3 articles

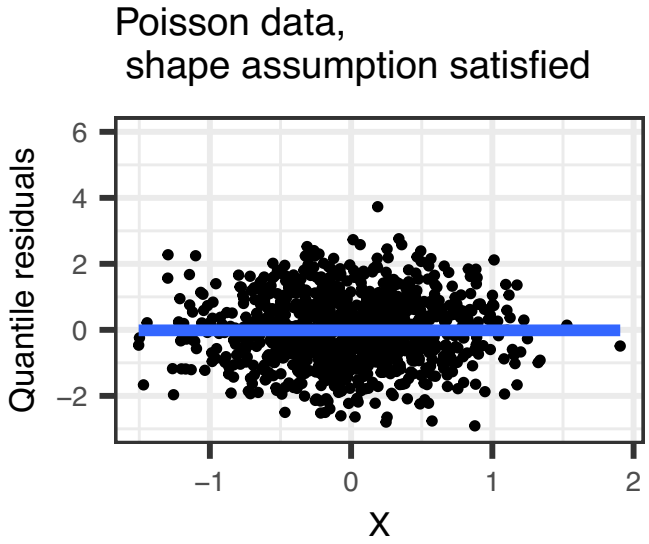
Model assumptions and diagnostics

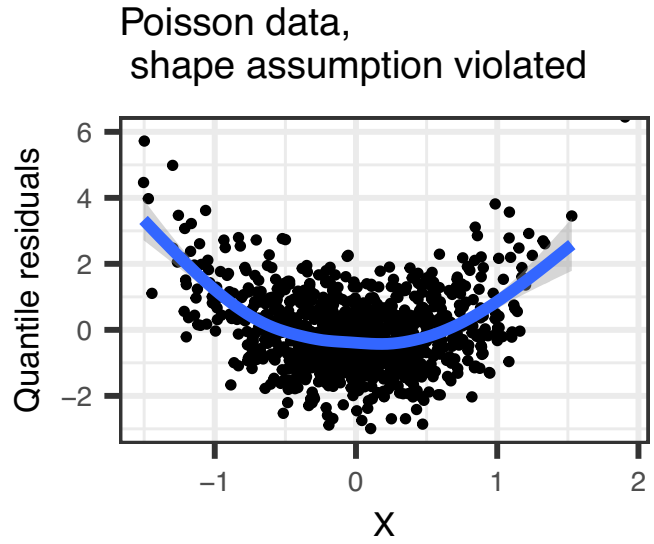
- Empirical log mean plots
- Quantile residual plots
- QQ plots of the quantile residuals
- Rootograms

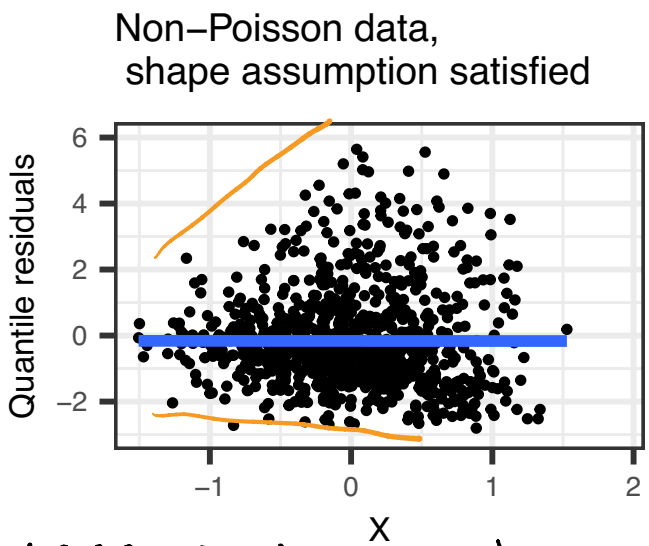
Activity

Work on the class activity (course website).

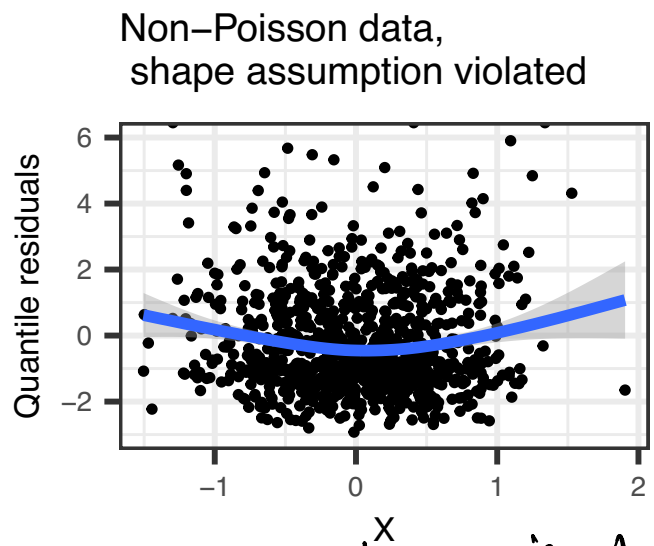
Class activity: Poisson regression assumptions







- non-constant variance
- variance is too large



- variance is too large
- AND trend to residuals

Using quantile residual plots

All assumptions are satisfied:

- Residuals are randomly scattered around 0, with no clear pattern
- Residuals have constant variance
- Residuals are mostly between -2 and 2

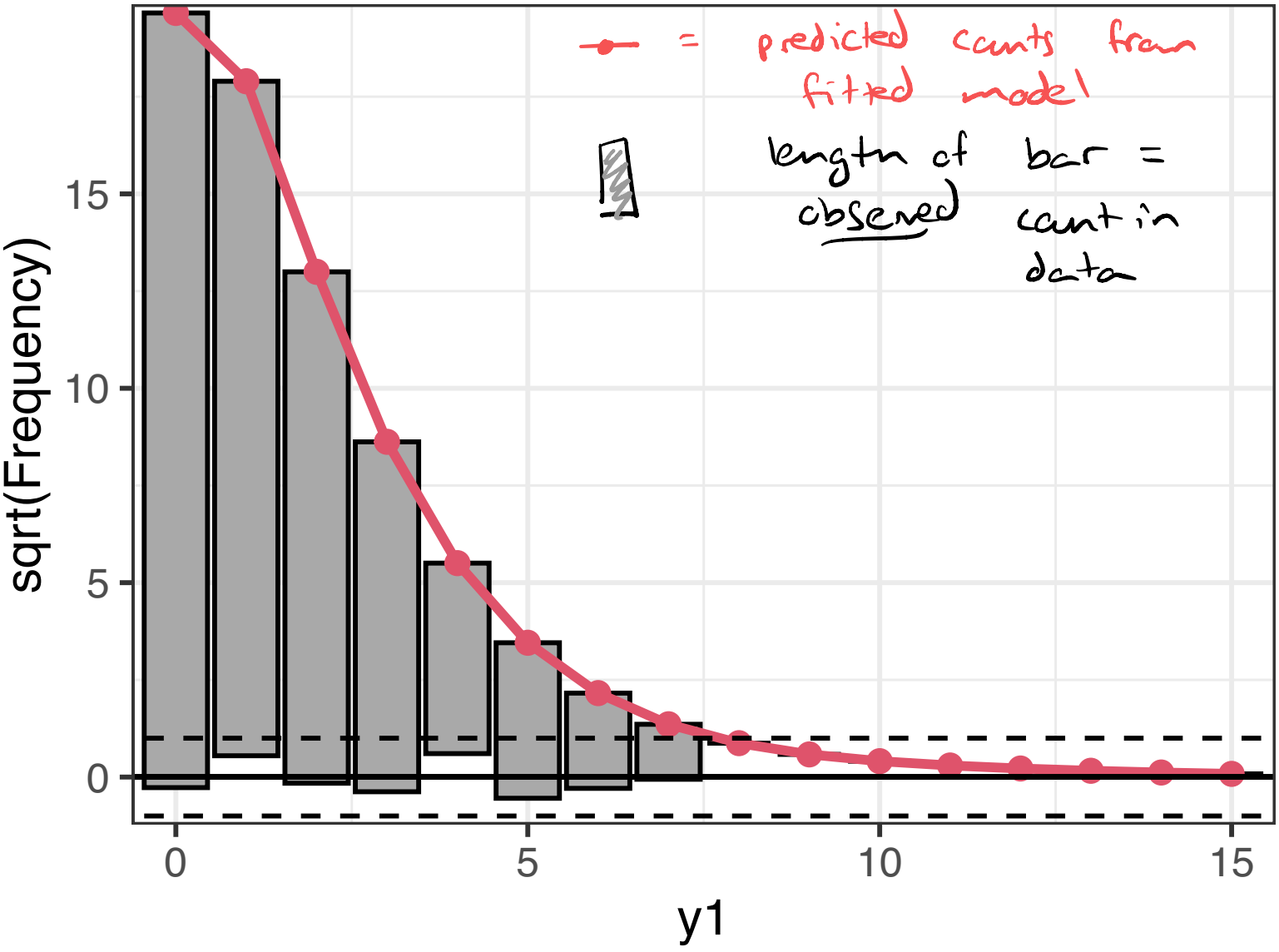
If you see a pattern in the residuals:

- Shape assumption may not be satisfied

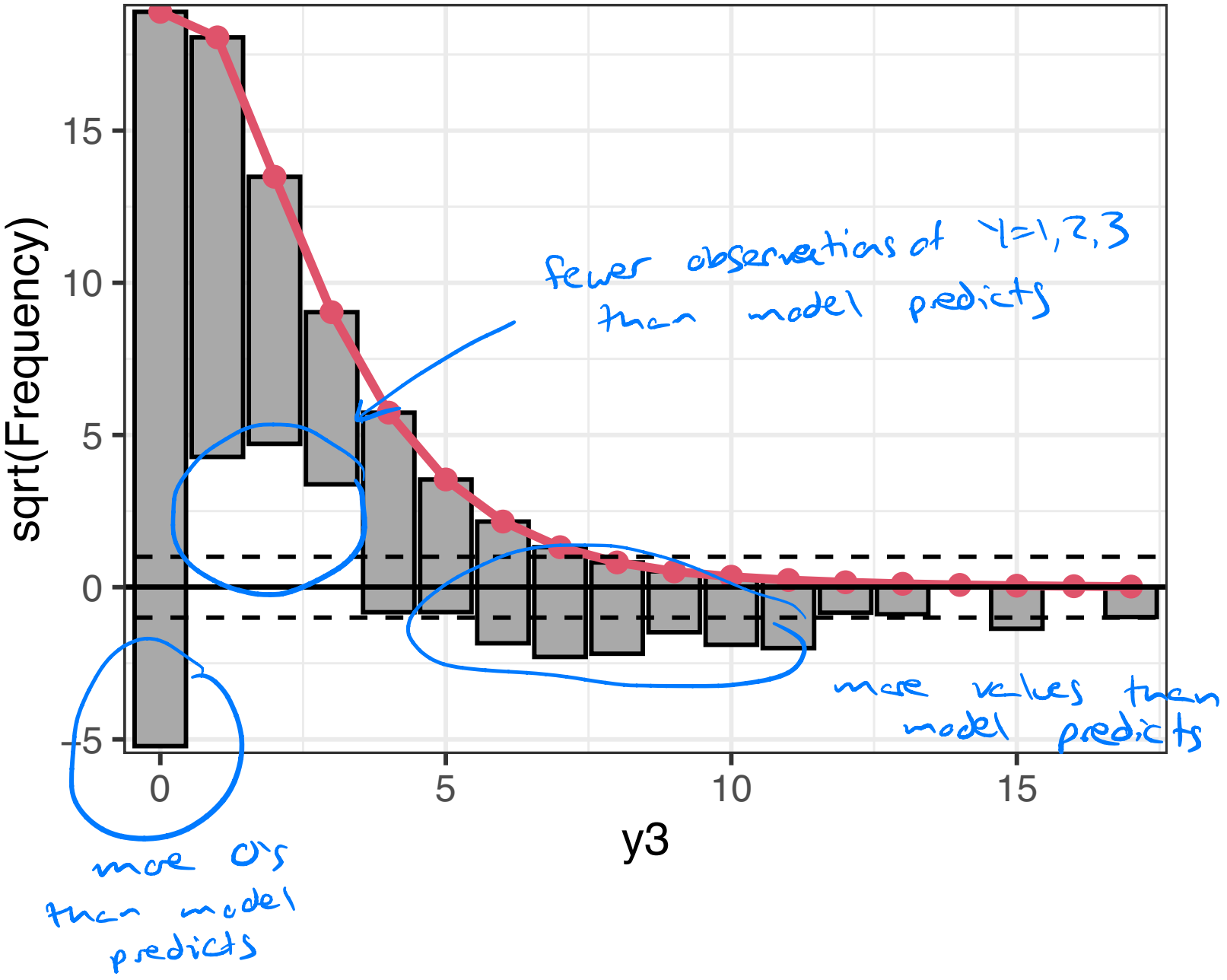
If residuals have non-constant variance, and/or variance is too large:

- Distribution assumption may not be satisfied

Rootograms

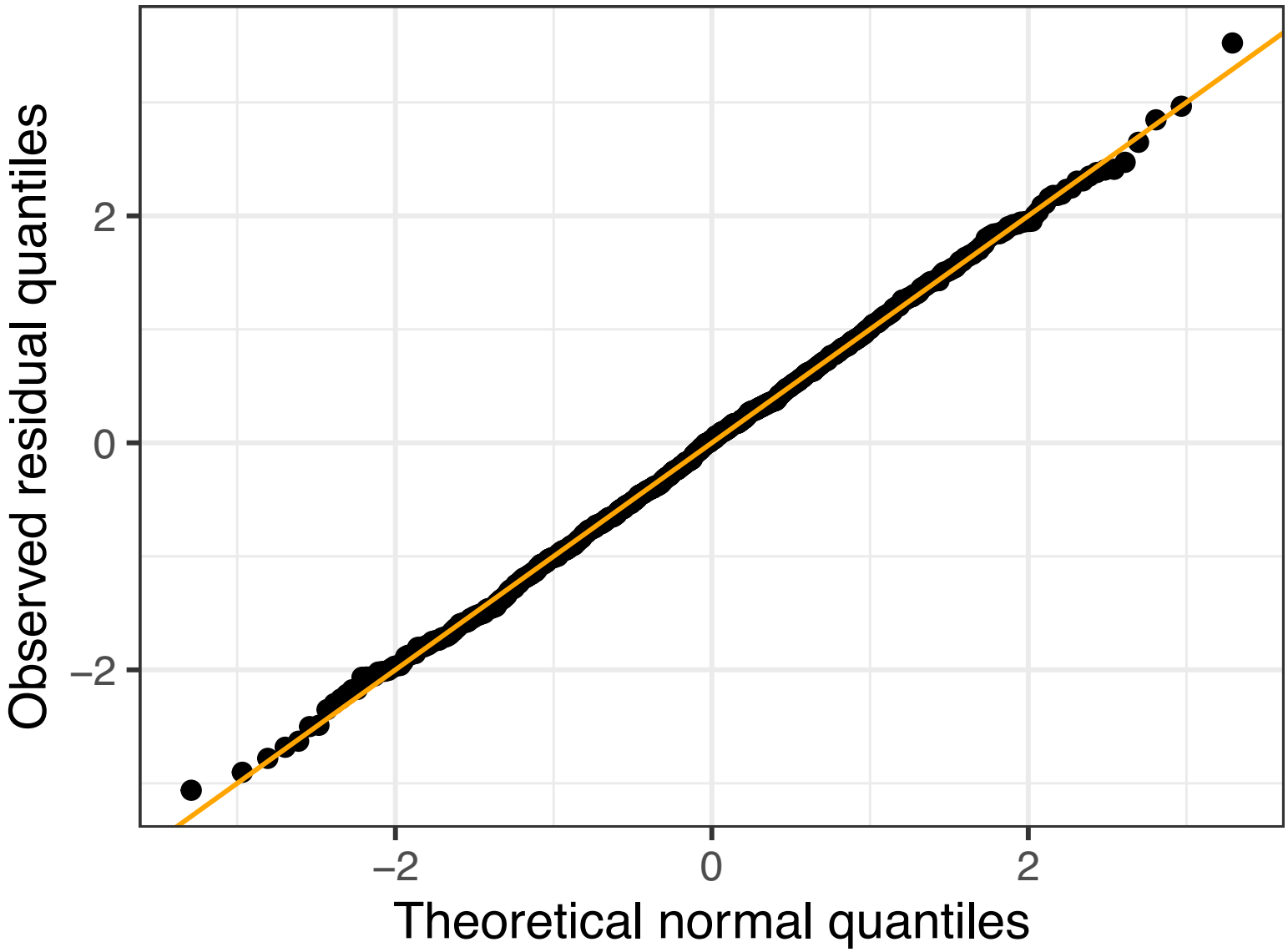

= predicted counts from fitted model
length of bar = observed count in data
sqrt(Frequency)
15
10
5
0
0
5
10
15
y1

Examining rootograms



QQ plots

When the model assumptions are satisfied:



QQ plots

When the model assumptions are not satisfied:

